

Teaching AI to Configure Proof Automation in Lean

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December 6, 2025

[Lean](#) is an open-source **theorem prover** and **programming language** that has become central to **AI for mathematics**, powering systems like AlphaProof, Harmonic's Aristotle, and dominating benchmarks like PutnamBench.

Challenge: **AI systems** are writing impressive Lean proofs, but **they often bypass powerful automation** tactics, **generating long manual proofs that a single tactic call could replace**.

- **The bottleneck is annotations:** metadata that tells automation how to use theorems.
- Most mathematicians can write proofs but can't write good annotations.
- **We propose training AI to predict annotations, trigger patterns, and instantiation constraints.**

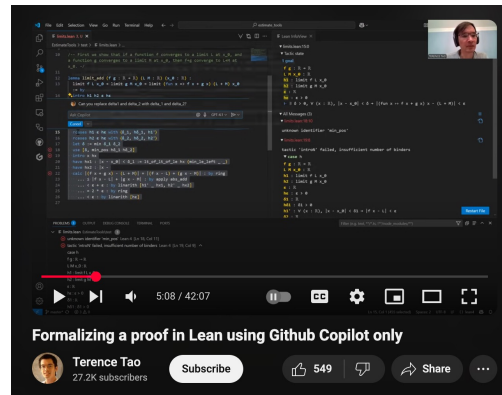
Lean is Taking Mathematics by Storm

*"Lean enables large-scale collaboration by allowing mathematicians to break down complex proofs into smaller, verifiable components. This formalization process ensures the correctness of proofs and facilitates contributions from a broader community. **With Lean, we are beginning to see how AI can accelerate the formalization of mathematics, opening up new possibilities for research.**" — Terence Tao*

Fermat's Last Theorem – Kevin Buzzard

Carleson's Theorem – Floris van Doorn

How did we get here?



Lean is Taking Mathematics by Storm

17 out of 29 AI for Math Fund winners reference Lean

Renaissance Philanthropy and XTX Markets Launch New \$9 million AI for Math Fund



2025 Winners

An AI-Focused Tactic for Language Learning
[Explore project](#)

A Dataset of Modern Formalized Theorem Statements
[Explore project](#)

A Principled Approach to Proof Search with Applications to Siderenko's Conjecture
[Explore project](#)

A Structured Representation of Tactics for Machine-Assisted Theorem Proving
[Explore project](#)

Bridging AI, Proof Assistants, and Mathematical Data (BRIDGE)
[Explore project](#)

Bridging Complexity and Automation to Advance Automatic Theorem Proving
[Explore project](#)

Bridging Proof and Computation
[Explore project](#)

Constraining LLMs for Theorem Proving
[Explore project](#)

Copilots for Isabelle
[Explore project](#)

Crowdsourcing and Reinventing the Next Generation of Dynamic and Scalable Math Benchmarks
[Explore project](#)

Databases of Structured Motivated Proofs
[Explore project](#)

DEEPER
[Explore project](#)

Document Level
Automatization

Domain Specific
Documentation for

Game Over or QED?

GNN SMT

renaissancephilanthropy.org/ai-for-math-fund-projects



The New York Times

Artificial Intelligence >

A.I.'s Math Problem

A.I. Training Data Disappears

Microsoft's Risk-Taker

Fine Print Changes

Quiz: Fake or Real Images?

Move Over, Mathematicians, Here Comes AlphaProof

A.I. is getting good at math — and might soon make a worthy collaborator for humans.

Share full article



47



Ring the gong at Google Deepmind's London headquarters, a ritual to celebrate each A.I. milestone, including its recent triumph of reasoning at the International Mathematical Olympiad. Google Deepmind

google-deepmind / formal-conjectures

<> Code

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formal-conjectures

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Fork 43

Star 478

main

Go to file

Code

About

Rekile and Paul-Lez

Fix: AMS codes (#185)

ed8a809 - yesterday

.devcontainer	feat: Add gitpod integration (#181)	2 days ago
.github	Fix caching issues with the doc buil...	last week
.vscode	vscode settings (#164)	5 days ago
FormalConjectures	Fix: AMS codes (#185)	yesterday
docbuild	Fix caching issues with the doc buil...	last week
scripts	ci: add a copyright header check (#...	2 weeks ago
.gitignore	move OpenProblems to third_party	2 months ago
.gitpod.yml	feat: Add gitpod integration (#181)	2 days ago
.mailmap	chore: add .mailmap (#60)	2 weeks ago

A collection of formalized statements of conjectures in Lean.

[google-deepmind.github.io/fo...](#)

[formal-mathematics](#)

[lean4](#)

Readme

Apache-2.0 license

Activity

Custom properties

478 stars

16 watching

43 forks

Report repository

AlphaProof & the International Math Olympiad

Determine all real numbers α such that, for every positive integer n , the integer

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \dots + \lfloor n\alpha \rfloor$$

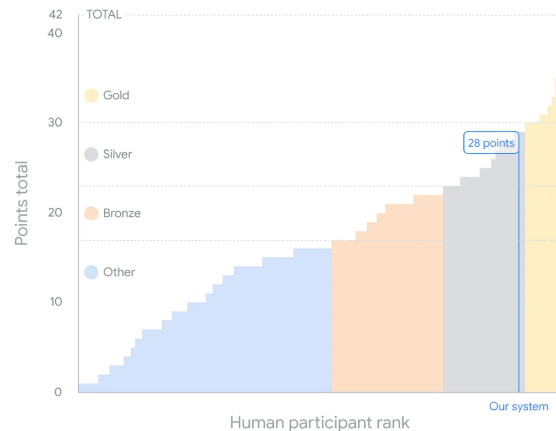
is a multiple of n . (Note that $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z . For example, $\lfloor -\pi \rfloor = -4$ and $\lfloor 2 \rfloor = \lfloor 2.9 \rfloor = 2$.)

Solution: α is an even integer.

```
open scoped BigOperators
```

```
theorem imo_2024_p1 :
  { (α : ℝ) | ∀ (n : ℕ), 0 < n → (n : ℤ) | (∑ i in Finset.Icc 1 n, ⌊ i * α ⌋) }
  = { α : ℝ | ∃ k : ℤ, Even k ∧ α = k } := by
  rw [(Set.Subset.antisymm_iff), (Set.subset_def), ]
  /- We introduce a variable that will be used
      in the second part of the proof (the hard direction)
```

Score on IMO 2024 problems



deepmind.google/discover/blog/ai-solves-imo-problems-at-silver-medal-level

IMO 2025: 3 Gold, 1 Silver*

OpenAI (informal)



Alexander Wei ✓
@alexwei_



1/N I'm excited to share that our latest @OpenAI experimental reasoning LLM has achieved a longstanding grand challenge in AI: gold medal-level performance on the world's most prestigious math competition—the International Math Olympiad (IMO).

Harmonic (Lean)

[Aristotle Achieves Gold Medal-Level Performance at the International Mathematical Olympiad, iOS App Beta Launch](#)

DeepMind (informal)

Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad

ByteDance (Lean)

ByteDance Seed Prover Achieves Silver Medal Score in IMO 2025



IMO-Level Proof Search as a Service

Aristotle Lean4 API

Mathematical Superintelligence, Formally Verified

Get access now

Ask follow-up...

+ Math Input

• Output: 7/16

```
have h_alg_int : IsIntegral (2 * Real.cos θ) <= by
  -- By Niven's theorem,  $2 \cos(\theta)$  is an algebraic integer.
  have h_alg_int : IsIntegral (Complex.exp (θ * Complex.I) + Complex.exp (-θ * Complex.I)) <= by
  by
    -- Since  $\theta$  is a rational multiple of  $\pi$ ,  $e^{i\theta}$  is a root of unity and hence an
    algebraic integer.
    obtain <= h1 : r : ℝ, 0 < r * Real.pi <= h0
    have h_root_of_unity : n : ℕ, n > 0 (Complex.exp (θ * Complex.I)^n = 1) <= h1
    -- Since  $\theta$  is rational, we can write  $\theta = \frac{2\pi k}{n}$  with  $k, n \in \mathbb{N}$ .
```

AlphaProof Interest Form

Thanks for your interest in AlphaProof. To deliver a reliable user experience, we are carefully managing our compute resources to maintain system stability. We will be reviewing applications and prioritizing access over time as we scale the system. We appreciate your patience and understanding.

To help you get the most from this process, we encourage you to first explore the [AlphaProof Blog Post](#), the [AlphaProof Nature Paper](#), and the [AlphaProof Tool Demos \[1\]\[2\]](#). Understanding AlphaProof beforehand will empower you to complete this form more effectively and confidently.

deepmind.google/alphaproof

aristotle.harmonic.fun

Highly Active, Healthy Competition on Putnam Bench

#	Model	num-solved	compute
1	Aleph Prover	500	pass@1, avg 1834 tool calls
2	Hilbert	462	avg pass@1840
3	Seed-Prover	329	MEDIUM
4	Ax-Prover ♥	91	pass@1, avg. 100 tool calls
5	Goedel-Prover-V2 ♥	86	pass@184
6	DeepSeek-Prover-V2 ♥	47	pass@1024
7	GPT-5 (ReAct, 10 turns)	28	pass@1, 10 tool calls
8	DSP+ ♥	23	pass@128
9	Bourbaki ♥	14	pass@512
10	Kimina-Prover-7B-Distill ♥	10	pass@192
11	Self-play Theorem Prover ♥	8	pass@3200
12	Goedel-Prover-SFT ♥	7	pass@512

UC San Diego/Apple, Sep 2s025

ByteDance, Jul 2025

Axiomatic_AI et al, Oct 2025

Princeton/NVIDIA et al, Aug 2025

DeepSeek-AI, Jul 2025

Microsoft Research et al, Jun 2025

Huawei et al, Jul 2025

Numina/Kimi, Apr 2025

Stanford, Mar 2025

Princeton et al, Apr 2025

trishullab.github.io/PutnamBench/leaderboard.html

♥ fully open-sourced

♥ partially open-sourced

Startups using Lean & AI



Math, Inc.



Logical Intelligence



...and more to come

Extensibility and Introspection

*"At Google DeepMind, we used Lean to build AlphaProof, a new reinforcement-learning based system for formal math reasoning. **Lean's extensibility and verification capabilities were key in enabling the development of AlphaProof.**" — Pushmeet Kohli, Vice President, Research Google DeepMind*

Lean 4 is implemented in Lean, allowing for unprecedented extension and introspection by users.

leanprover-community/repl is a simple Lean program for communicating with the Lean compiler via JSON.

AI labs have customized it in many ways, especially around parallelized proof tree search.

stanford-centaur/PyPantograph is the state of the art for open-source Lean REPLs.

Auto-formalization

Auguste Poiroux

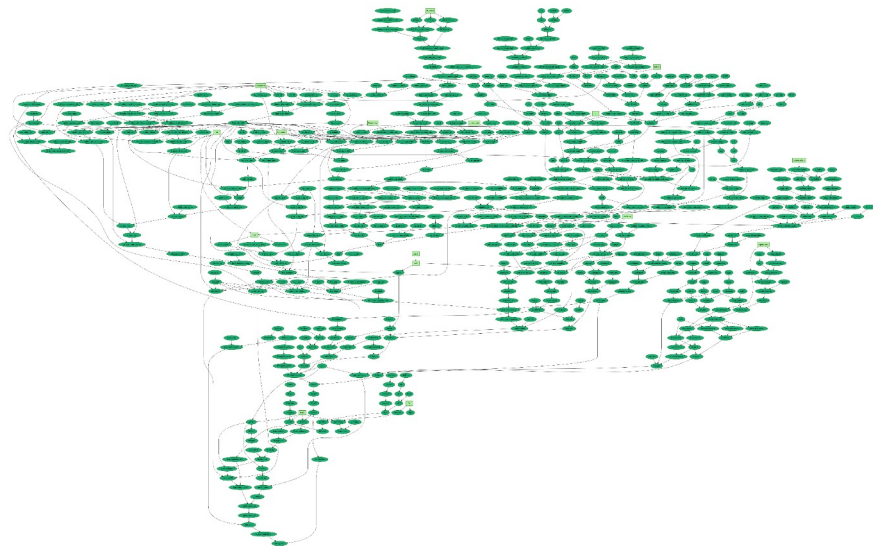
Dear all,

We, at Math, Inc., are excited to introduce [Gauss](#), a new autoformalization agent designed to assist mathematicians in their Lean formalization.

Using Gauss, and building over the recent [medium PNT progress](#) by Kontorovich et al., we formalized the **strong Prime Number Theorem** in Lean. Over the course of three weeks, Gauss produced ~25k lines of Lean code and over 1,000 theorems and definitions. To put these numbers in perspective, our previous iteration, three months ago, produced ~3.5k lines of Lean code and 100 theorems and definitions to formalize the [abc conjecture almost always](#).

Our goal is to make Gauss accessible and easy to use. If you are interested, you can register for early access to Gauss [here](#). We are happy to discuss about Gauss and Math, Inc. in the discussion thread.

github.com/math-inc/strongpnt



Auto-formalization Inspection

```

have h := zetacnot0 t
exact norm_pos_iff.mpr h

lemma Zeta1_Zeta_Expand:
  ... ∃ A > 1, ∃ b > 1,
  ... ∀ (t : ℝ) · (ht : |t| > 2)
  ... (r1 · r · R1 · R : ℝ)
  ... (hr1_pos : 0 < r1) · (hr1_lt_r : r1 < r)
  ... (hr_pos : 0 < r) · (hr_lt_R1 : r < R1) · (hr1_pos : 0 < R1) · (hr1_lt_
  ... let c := (3/2 : ℂ) + I * t;
  ... ∀ (hfin : (zerosetKfRc · R1 · c · riemannZeta).Finite),
  ... ∀ z ∈ closedBall c r1 \ zerosetKfRc · R1 · c · riemannZeta,
  ... ||logDerivZeta z - ∑ p ∈ hfin.toFinset,
  ... ((analyticOrderAt · riemannZeta p).toNat : ℂ) / (z - p)|| ≤
  ... (16 * r^2 / ((r - r1)^3) +
  ... 1 / ((R^2 / R1 - R1) * Real.log (R / R1))) * (Real.log |t| + Rea
  -- Apply the three lemmas mentioned in the informal proof
obtain ⟨b, hbgt1, hb⟩ := zeta32upper
obtain ⟨A, hAgt1, hA⟩ := zeta32lower_log

-- Provide the constants A, b as required
refine ⟨A, hAgt1, b, hbgt1, ?_⟩
intro t ht r1 r R1 R hr1_pos hr1_lt_r hr_pos hr_lt_R1 hr1_pos hr1_
intro c hfin z hz

-- Apply log_Deriv_Expansion_Zeta
have hexp_lemma := log_Deriv_Expansion_Zeta t ht r1 r R1 R hr1_pos

-- Set B = b * |t| as mentioned in informal proof
have htpos : (0 : ℝ) < |t| := by linarith [ht]
have hbgt1 : b * |t| > 1 := by
  have hb_pos : (0 : ℝ) < b := by linarith [hbgt1]

```

Mathematical significance:

This is likely part of a proof related to the Prime Number Theorem, where you need precise control over how well the logarithmic derivative of the zeta function can be approximated by summing over its zeros. The bound includes both geometric terms (involving the radii) and growth terms (logarithmic in $|t|$).

The proof strategy mentioned in the comments indicates it combines results from `zeta32upper` and `zeta32lower_log` lemmas to establish the required constants and bounds.

What is the definition of the logarithmic derivative here?

PNT3_RiemannZeta.lean:4428-4440

I'll search for the definition of `logDerivZeta` in this file to show you how the logarithmic derivative is defined here.

Searched text for `logDerivZeta`
> (`**/StrongPNT/PNT3_RiemannZeta.lean`), 20 results

Perfect! I found the definition. Let me read the context around line 3495 to give you the complete picture:

✓ Read `PNT3_RiemannZeta.lean`, lines 3490 to 3500

Vibe Proving

Forbidden Sidon subsets of perfect difference sets,
featuring a human-assisted proof

Boris Alexeev

ChatGPT*

Lean[†]

Dustin G. Mixon^{‡§}

Abstract

We resolve a \$1000 Erdős prize problem, complete with formal verification generated by a large language model.

In over a dozen papers, beginning in 1976 and spanning two decades, Paul Erdős repeatedly posed one of his “favourite” conjectures: every finite Sidon set can be extended to a finite perfect difference set. We establish that $\{1, 2, 4, 8, 13\}$ is a counterexample to this conjecture.

During the preparation of this paper, we discovered that although this problem was presumed to be open for half a century, Marshall Hall, Jr. published a different counterexample three decades *before* Erdős first posed the problem. With a healthy skepticism of this apparent oversight, and out of an abundance of caution, we used ChatGPT to vibe code a Lean proof of both Hall’s and our counterexamples.



Ø ...



Aristotle from @HarmonicMath just proved Erdos Problem #124 in @leanprover, all by itself. This problem has been open for nearly 30 ...

Q 17

↑↱ 40

♡ 561

89K



@logic_int



4 math problems unsolved for 20+ years. Formal proofs in Lean 4. Less than 48 hours. Under \$5k total.

- ✓ Binomial tail bounds conjecture (Telgarsky, 2009)
- ✓ Quantum gate lattice approximation (Greene & Damelin, 2015)*
- ✓ Erdős 124
- ✓ Erdős 481
- ✓ #1 on PutnamBench leaderboard

The era of AI mathematics is here.



Aristotle by @Harmonic is acing group-theory puzzles.

Here is a complete formal proof of the popular Yu Tsumura 554 puzzle. What's nice is that the proof is very transparent, with easy-to-follow steps. It was generated in less than an hour without any hints. I am ...

[Show more](#)

Carina Hong  @CarinaLHong · Dec 2

AxiomProver solved Erdos problem #481 - took 5 hours

#124 simplified version took over 24 hours (oof) and was not as succinct as we'd like it to be

```
rw ldn_of_nthdiv_h_ratio_extval at hbound
  |> (bound : k / m * k / (12 * M / c + 1))

-- From k / m * k / (12 * M / c + 1), get m^k / k + 2 * c * k + 1 + 2 * c * k
have h_ratio_pos : (k : R) / m ^ k / (12 * M / c + 1) <= div_pos h_bound h_ext_pos
have h_ratio_bound : m ^ k / (k : R) + 2 * M / c <= 10
have h1 : c <= m ^ k / (k : R) + 1 / (k : R) / m ^ k <= by
  rw [lme_div, lme_div1]
  rw [h2]
  rw [h3]
  rw [h4]
  rw [h5]
  rw [h6]
  rw [h7]
  rw [h8]
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  rw [h416]
  rw [h417]
  rw [h418]
  rw [h419]
  rw [h420]
  rw [h421]
  rw [h422]
  rw [h423]
  rw [h424]
  rw [h425]
  rw [h426]
  rw [h427]
  rw [h428]
  rw [h429]
  rw [h430]
  rw [h431]
  rw [h432]
  rw [h433]
  rw [h434]
  rw [h435]
  rw [h436]
  rw [h437]
  rw [h438]
  rw [h439]
  rw [h440]
  rw [h441]
  rw [h442]
  rw [h443]
  rw [h444]
  rw [h445]
  rw [h
```

Autonomous Exploration

MATHEMATICAL EXPLORATION AND DISCOVERY AT SCALE

BOGDAN GEORGIEV, JAVIER GÓMEZ-SERRANO, TERENCE TAO, AND ADAM ZSOLT WAGNER

ABSTRACT. AlphaEvolve [223] is a generic evolutionary coding agent that combines the generative capabilities of LLMs with automated evaluation in an iterative evolutionary framework that proposes, tests, and refines algorithmic solutions to challenging scientific and practical problems. In this paper we showcase AlphaEvolve as a tool for autonomously discovering novel mathematical constructions and advancing our understanding of long-standing open problems.

1.5. Building a pipeline of several AI tools. Even more strikingly, for the finite field Kakeya problem (cf. Problem 6.1), AlphaEvolve discovered an interesting general construction. When we fed this programmatic solution to the agent called Deep Think [148], it successfully derived a proof of its correctness and a closed-form formula for its size. This proof was then fully formalized in the Lean proof assistant using another AI tool, AlphaProof [147]. This workflow, combining pattern discovery (AlphaEvolve), symbolic proof generation (Deep Think), and formal verification (AlphaProof), serves as a concrete example of how specialized AI systems can be integrated. It suggests a future potential methodology where a combination of AI tools can assist in the process of moving from an empirically observed pattern (suggested by the model) to a formally verified mathematical result, fully automated or semi-automated.

New result distribution

Visualization of results across 67 problems.

■ New result
 ■ Former new result, got improved upon
 ■ Worse than literature bound
■ Matched known optimal bound
 ■ Matched literature bound / N/A



Autonomous Exploration

```

83  -- Inside this block, when we write `(K1 p)`, Lean uses the `p` from the context.
84  lemma card_K1 (hp_odd : p > 2) : (4 * (K1 p).card : ℤ) = p^3 + 2*p^2 + p := by
85    have := (Fact.out : (p.Prime)).odd_of_ne_two fun and' => by simp_all
86    rcases this with ⟨x, hx⟩
87    subst hx
88    delta K1
89    rw_mod_cast[Finset.card_filter]
90    trans(4)*ΣS:ZMod (2*x+1) ×_ ×_, ite (IsSquare (S.1^2+4* S.2.1) ∧ IsSquare (S.1^2+4* S.2.2)) (1) 0
91    · exact (congr_arg _) (Fintype.sum_equiv ⟨ fun and=> (and 0, and (1), and 2), fun and=>![_,_,_]
92    · trans(4)*ΣS:ZMod (2*x+1), (( Finset.univ.image fun and=> (and* and-S^2 :) /4) ×s.image (@ fun a
93    · rw[ Fintype.sum_prod_type, Fintype.sum_congr _ _ fun and=>(Finset.card_filter _ _).symm.trans
94    field_simp[IsSquare, sub_eq_iff_eq_add'.trans (comm), show (4 :ZMod (2*x + 1))#0 from(( ZMod.
95    · push_cast[sq, add_assoc, .>., Finset.card_product, true, Finset.mul_sum]at *
96    transΣS:ZMod (2*x+1), 4*(( Finset.univ.filter fun and=>and.val≤and.val).image fun p=>(p*p-S
97    · push_cast only [le_refl, true, Finset.filter_true_of_mem, implies_true, sq]
98    · transΣS:ZMod (2*x+1), 4*(( Finset.range (x + 1)).image fun p : ℕ=>(p*p-S* S:ZMod (2*x + 1)
99    · refine Fintype.sum_congr _ _ fun and=>congr_arg _ ((congr_arg (.^2) (congr_arg _ (Finset.
100    use Finset.mem_image.2 (if h: a.val≤x then⟨ _, Finset.mem_range_succ_iff.2 h, by simp_aritl
101    · rewrite[ Finset.sum_congr rfl fun and x =>(congr_arg _) ((congr_arg) ( .^2) ( Finset.card.
102    · simp_all[show (2*x+1)*(4*(x+1)^2) = (2*x+1)^3+ (2 *((2*x+1)*(2*x + 1))+(2*x + 1))by rii
103    · field_simp[show (4 :ZMod (2 *by bound + 1))#0 from((ZMod.isUnit_iff_coprime _ _).2 (Odd
104    use (by valid:).elim (by valid . (ZMod.val_cast_of_lt (by valid)).symm.trans.comp (. ▶ Z

```



A vibrant community of users at leanprover.zulipchat.com

 **Machine Learning for Theorem Proving** Machine Learning and AI or theorem proving. HOList, AI for...

Standard view

Filter

MCP Tools for LLMs and Agentic Mathematics
Building an Autoformalizer on Analysis
✓ Executing Conv and Calc in Pantograph
Dataset to rule them all?
REPL: automated incremental state reuse across commands
Better way for tracing tactic states
Blind Speculation about IMO 2025
Claude 4 agent in VS Code
DeepMind and Navier Stokes
LeanTool feature: Sorry Hammer
AI tutorial for LEAN beginner
Proof or Bluff
Autoformalization of the probabilistic abc-conjecture
Illusion of Thinking
Tool for Lean code verify
Gemini 2.5 Pro 06/05
Current state of AI for Mathematics and what could come n...
System card: Claude Opus 4 & Claude Sonnet 4
DeepSeek-Prover V2
I was pleasantly surprised by DeepSeek

Prospective: AI+CS as the Next Frontier is Imminent

[leanprover/cslib](#) founded, supported by AWS, Google DeepMind, SDU, Centaur (Stanford)

Companies are starting to pivot their math-proven AIs towards program verification

► Beyond Math

While the IMO benchmark highlights Aristotle's mathematical ability, the implications reach much further. The same architecture that can solve Olympiad problems also supports:

- Code generation that doesn't require human verification
- Engineering tasks where precision is mission-critical
- Scientific research where progress depends on rigorous mathematical understanding

[harmonic.fun/news#blog-post-aristotle-tech-report](#)

Spec auto-formalization + program synthesis + verification as a service



Ilya Sergey
@ilyasergey



The proof of the last remaining Lean theorem in our upcoming conference submission has now been completed with the help of AI.

The research community's perception of program verification is about to change irreversibly.

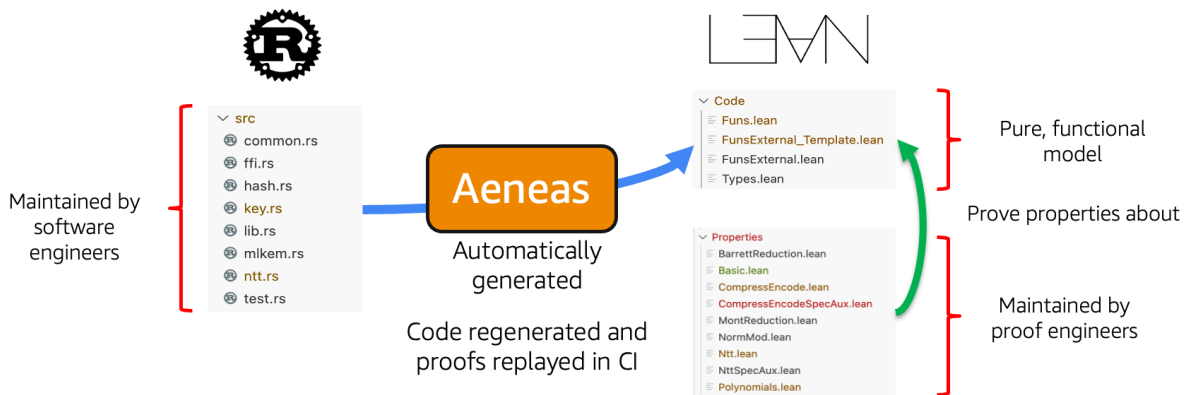
8:32 PM · Nov 13, 2025 · **15.8K** Views

Prospective: AI+CS as the Next Frontier is Imminent



Beyond math: Aristotle achieves SOTA 96.8% proof generation on VERINA: Benchmarking Verifiable Code Generation. You can read more about this performance on our engineering blog linked in bio

12:24 PM · Dec 3, 2025 · 1,860 Views



Teaching AI to Configure Proof Automation in Lean

NeuroSymbolic AI

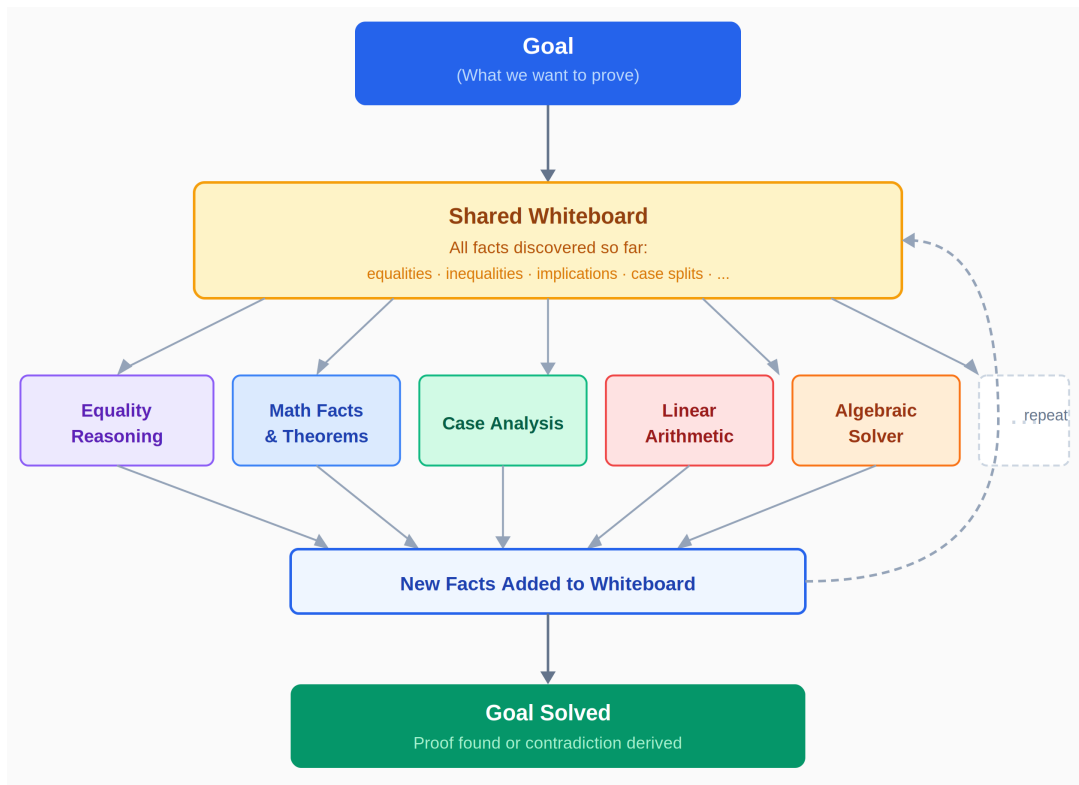
The future of AI for mathematics is not pure neural or pure symbolic. It's the combination.

Symbolic engines provide exhaustive search in structured domains, guaranteed correctness, and decision procedures refined over decades.

AI provides pattern recognition, heuristic guidance, and retrieval. The question is how to combine them effectively.

grind is the symbolic in Neurosymbolic AI

New proof automation (Lean v4.22 – released mid August) developed by Kim Morrison and myself.



Automating Quantum Algebra

```
example {α} [CommRing α] [IsCharP α 0] (d t c : α) (d_inv PS03_inv : α)
  (Δ40 : d^2 * (d + t - d * t - 2) * (d + t + d * t) = 0)
  (Δ41 : -d^4 * (d + t - d * t - 2) *
    (2 * d + 2 * d * t - 4 * d * t^2 + 2 * d * t^4 + 2 * d^2 * t^4 - c * (d + t + d * t)) = 0)
  : d * d_inv = 1 →
    (d + t - d * t - 2) * PS03_inv = 1 →
    t^2 = t + 1 := by
grind
```

This is not a toy: it encodes a real algebraic constraint derived from relations among diagrams in a pivotal tensor category.

grind can handle this kind of reasoning automatically, in milliseconds.

Generate proof term is massive, more than 20,000 terms.

grind annotations

```
@[grind =]
theorem length_tail {α} {as : List α} : as.tail.length = as.length - 1 := by
  cases as <;> rfl

@[grind →]
theorem mem_or_eq_of_mem_set {α} {as : List α} {i : Nat} {a b : α} : a ∈ as.set i b → a ∈ as ∨ a = b
```

Lean standard library comes pre-annotated.

We are annotating Mathlib, it is a monumental task. What about user libraries?

Can AI make theorem-proving AI better?

If we had perfect annotations:

Proof synthesis becomes dramatically easier (predict grind instead of 20 tactic steps)

AI-generated proofs become shorter, more robust, more maintainable

Another reward signal: #grind_lint

```

/--
info: instantiating `List.replicate_sublist_iff` triggers 30 additional `grind` theorem instantiations
---
info: List.replicate_sublist_iff
[thm] instances
  [thm] List.count_eq_length_filter ↦ 3
  [thm] List.count_le_length ↦ 3
  [thm] List.length_filter_le ↦ 3
  [thm] List.Sublist.count_le ↦ 3
  [thm] List.Sublist.eq_of_length ↦ 3
  [thm] List.Sublist.length_le ↦ 3
  [thm] List.replicate_sublist_iff ↦ 2
  [thm] List.subset_def ↦ 2
  [thm] List.Sublist.subset ↦ 2
  [thm] List.count_replicate ↦ 1
  [thm] List.filter_replicate ↦ 1
  [thm] List.filter_subset ↦ 1
  [thm] List.length_replicate ↦ 1
  [thm] List.sublist_filter_iff ↦ 1
  [thm] List.Sublist.filter ↦ 1
---
info: Try this to display the actual theorem instances:
  [apply] set_option trace.grind.ematch.instance true in
    #grind_lint inspect (detailed := 20) List.replicate_sublist_iff
-/
#guard_msgs in
#grind_lint inspect (detailed := 20) List.replicate_sublist_iff

#grind_lint check (min := 30) in List

```

A tool for analyzing annotations.

We want to minimize the number of theorem instantiations generated by grind without reducing the number of problems we can solve.

What about unannotated libraries?

```

301 /-- A product set is included in a product set if and only factors are included, or a factor
302 first set is empty. -/
303 theorem prod_subset_prod_iff : s ×s t ⊆ s1 ×s t1 ↔ s ⊆ s1 ∧ t ⊆ t1 ∨ s = ∅ ∨ t = ∅ := by
304   in cases (s ×s t).eq_empty_or_nonempty with h | h
305   · simp [h, prod_eq_empty_iff.1 h]
306   have st : s.Nonempty ∧ t.Nonempty := by rwa [prod_nonempty_iff] at h
307   refine (fun H => Or.inl (?, ?), ?)
308   · have := image_mono (f := Prod.fst) H
309     rwa [fst_image_prod _ st.2, fst_image_prod _ (h.mono H).snd] at this
310   · have := image_mono (f := Prod.snd) H
311     rwa [snd_image_prod st.1, snd_image_prod (h.mono H).fst] at this
312   · intro H
313     simp only [st.1.ne_empty, st.2.ne_empty, or_false] at H
314     exact prod_mono H.1 H.2
315

```

Lean InfoView

▼ Prod.lean:310:27

▼ Tactic state

1 goal

▼ case inr.refine_2

α : Type u₁

β : Type u₂

s s₁ : Set α

t t₁ : Set β

h : (s ×^s t).Nonempty

st : s.Nonempty ∧ t.Nonempty

H : s ×^s t ⊆ s₁ ×^s t₁

this : Prod.snd '' s ×^s t ⊆ Prod.snd '' s₁ ×^s t₁

grind +suggestions is extensible

You can attach your own AI.

Lean comes with a symbolic suggestion engine.

You can apply it to all proofs in huge math libraries and learn which theorems are the most effective.

Already available in our master branch.

Coming at v4.27.0

grind has an interactive mode

```
example (m : IndexMap  $\alpha$   $\beta$ ) (a a' :  $\alpha$ ) (b :  $\beta$ ) :  
  a'  $\in$  m.insert a b  $\leftrightarrow$  a' = a  $\vee$  a'  $\in$  m := by  
  grind?
```

Already available in v4.25.0. Higher level of abstraction.

Prediction: AI will learn how to use grind interactive mode in a few months.

What do we need from you?

Baselines: How well does Claude / Gemini / GPT-5 do zero-shot on annotation prediction?

RL experiments: Can you train an agent with #grind_lint as reward that learns to write good constraints?

Suggestion oracle: Can you build a neural model that, plugged into grind +suggestions, improves success rate on held-out goals?

The hard version: Given a new library the model hasn't seen, can it produce annotations matching expert quality?

Before we wrap up...



Lean FRO: Shaping the Future of Lean Development

The Lean Focused Research Organization (FRO) is a non-profit dedicated to Lean's development.

Founded in **August 2023**, the organization has 19 members.

Its mission is to enhance critical areas: **scalability, usability, documentation**, and **proof automation**.

It must reach **self-sustainability in August 2028** and become the **Lean Foundation**.

We are very grateful for all philanthropic support we have received.

Lean FRO: by numbers

25 releases and **+7,000 pull requests** merged in the main repository only since its launch in July 2023.

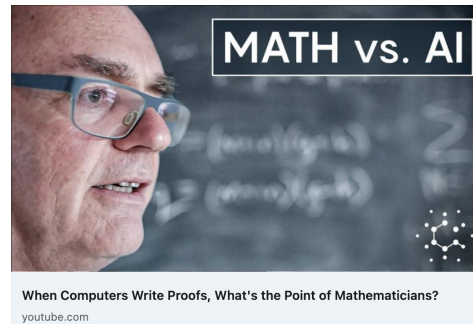
Public roadmaps: <https://lean-lang.org/fro/roadmap/y3/>

Lean project was featured in multiple venues NY Times, Quanta, Scientific American, etc.



A.I. Is Coming for Mathematics, Too

For thousands of years, mathematicians have adapted to the latest advances in logic and reasoning. Are they ready for artificial intelligence?





How can I contribute?

Help building [Mathlib](#) and [CSLib](#).

Want to engage with the vibrant Lean community? Join our [Zulip channel](#).

Start your own open-source Lean project! Your package will be available on our registry [Reservoir](#).

Start using Lean online: live.lean-lang.org

Support the Lean FRO: Funding, partnerships, or simply advocating the project.

Conclusion

Lean is an **efficient programming language** and **proof assistant**.

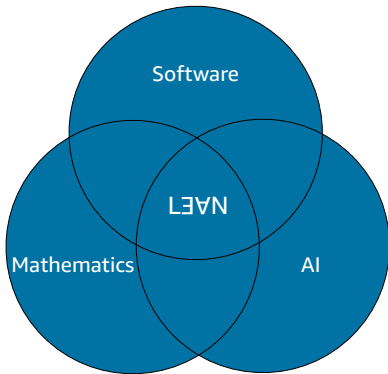
Lean enables Verified AI for Mathematics and Code.

AI should learn to use new symbolic tools.

Proof automation based on NeuroSymbolic AI will accelerate progress.

We have the infrastructure. We need you to build the AI.

[Teaching AI to Make Proof Automation Work](#)



Thank You

<https://leanprover.zulipchat.com/>

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#leanlang, #leanprover

<https://lean-lang.org/>

